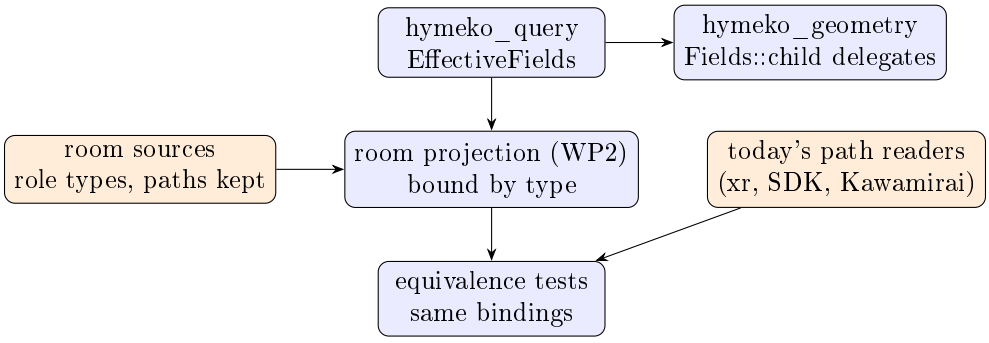
# Declarative room structure (P1a)

Created-at: 2026-09-27 02:45 +02:00. Authorization 2026-09-27 (after 02:13): P1a now, inherited-field resolution in `hymeko_query`; (after 02:20): commit the uncommitted Transformation and SDK state first as separate prior commits (done: `transformation` dc002b2, `hymeko_web` da94b0b, both on `feat/declarative-room`). Standing rule: commit and push, new feat branch per repository. Predicates, declared conditions and HySTL stay with the RL line.

**ETA** WP1 1–1.5 h (until about 04:15). WP2–WP4 3–5 h after the vocabulary decision below. Basis: WP1 is one extracted reader, one delegation and focused tests; WP2 is a projection of the size of `SpatialQuery` plus two model sources and equivalence tests.

### Findings that shape the plan

1. The own-then-bases field reader exists once, crate-private: `hymeko_geometry::kinematics::actuation::Fields` (own declaration first, then bases in authored order, cycle-safe). `hymeko_webxr`'s `attr_value`, `hymeko_query`'s `MatchContext::get_field` and the SDK inspector read direct children only; inheritance tests exist in three places (`decl_inherits`, `isa_transitive`, `check_inherits_simple`).
2. Two marker vocabularies coexist: ROS (`ros.marker`: `dictionary`, `marker_id`, `size_m`; `ros.registration`) used by the deployed room, and the spatial one (`spatial.kinds.fiducial_anchor`: `family`, `identifier`, `size_m`; `@fiducial_registration`) with a type-bound query (`hymeko_webxr::SpatialQuery`).
3. The ecosystem map lists `hymeko_webxr` as used by `hymeko_quest`; no XR, SDK, Kawamirai or HIVE source uses it (finding: map and code disagree).
4. `hymeko_query`'s integration target includes `bench_workflow`, an un-ignored timing `#[test]` (finding; not changed by this plan).




### Work packages

WP1 `hymeko_query`: public effective-field reader (own declaration, then bases in authored order, cycle-safe, bounded), with a dotted-path form on `MatchContext` that is opt-in (the existing `get_field` keeps its direct-child semantics, so template emitters do not change output). `hymeko_geometry::Fields::child` delegates to it.

WP2 Room projection (*decision pending*): typed views of room root marker, landmarks, tracked headset, robot marker and editable room objects, bound by type through queries, never by path. Option A: extend `hymeko_webxr` (spatial owner per ecosystem map and AGENTS.md) and align the ROS marker with `fiducial_anchor`. Option B: room vocabulary beside the ROS one, projection in the SDK next to `RoomLayout`.

WP3 Model sources: role types on the deployed room (paths unchanged) and a typed rewrite of `hive-room.hymeko`; CLI gate for each file.

WP4 Equivalence tests: for both sources, today's path-based readers and the new projection yield the same bindings; README, symbols, duplication, report; commits and push per repository.

**Affected files (WP1)** `hymeko_query/src/{lib.rs, fields.rs (new), rewrite/match_context.rs}`, `hymeko_geometry/src/kinematics/actuation.rs`, tests in both crates, `SYMBOLS.md`. WP2–WP4 files are listed in the amendment after the decision. **CORE.YAML items touched:** none (Transformation declares none; Core is not edited).

**Interface changes (WP1)** New `hymeko_query::EffectiveFields` (child, value); `MatchContext::effective_fiel` No existing signature changes.

**Amendment 2026-09-27 02:47 +02:00** A separate `EffectiveFields` struct repeated `QueryEngine`'s `ir`/`resolver` holder (token clone, +2/+16 against dc002b2). The reader is therefore `QueryEngine::effective_child( key)` and `effective_value(decl, key)` (module `fields.rs`); `QueryEngine::ir()` and `resolver()` now return the engine's `'a` borrow, so a value outlives a temporary engine (widening, source-compatible for callers). Duplication back to 384 clones / 6053 lines.

**Tests** Unit: own value overrides a base; bases in authored order; multi-level inheritance; diamond and cycle terminate; missing field; dotted path through an inherited block. Regression: `hymeko_geometry` inherited-material/geometry and actuation tests unchanged. Correctness only; `--skip bench_` for the query integration target; no measurement.

**Budgets** Reader cost is bounded by the base graph (visited set); no allocation per direct hit beyond the walk stack. No performance measurement is authorized.

**Rollback** Revert the WP1 commit; geometry returns to its private reader.

**Risks** (1) Changing `get_field` globally would change emitted URDF/SDF; the new reader is opt-in. (2) The vocabulary decision changes a deployed model (WP3); live HIVE room migration is P2 and not part of this plan. (3) Dirty sibling state is now committed but still depends on uncommitted Core APIs; builds are local integration builds.